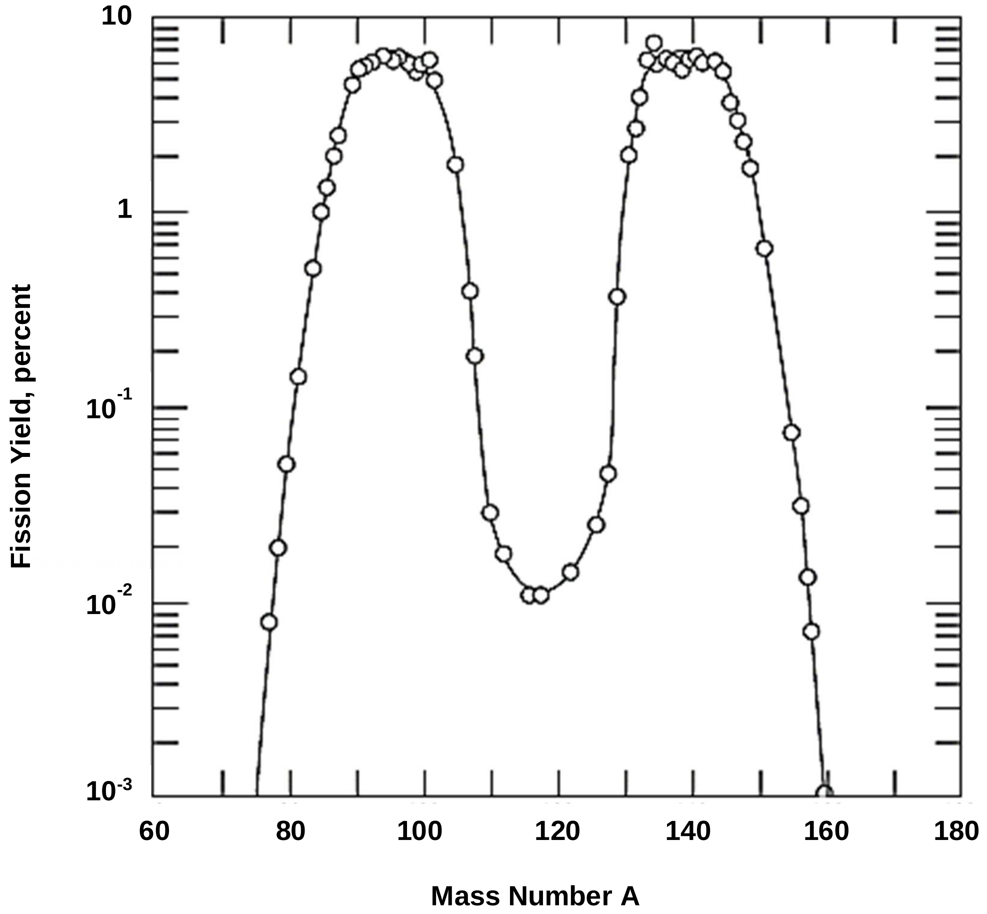
7 When a very strong earthquake of magnitude 9.0 hit Japan on 11 March 2011, the accompanying tsunami caused the devastation of the Fukushima Dai-ichi Nuclear Power Plant (Fk-1). A large amount of radioactive material leaked out of Fk-1 as a result, mainly through cracks made by the quake on some reactors, explosions, intentional vents to relieve pressure, and leakage of cooling water that was contaminated with the melted fuel rod debris.

Fk-1 was capable of generating up to 4.7 gigawatts of power through neutron-induced fissions of Uranium-235 (^{235}U) nuclei. Fission resulting from neutron absorption is called *induced* fission. Some nuclides can also undergo *spontaneous* fission, but that is quite rare. Nuclear fission is a decay process in which an unstable nucleus splits into two fragments of comparable mass, although nearly equal mass is unlikely. Over 100 different nuclides, representing more than 20 different elements, have been found among the fission products. Fig. 7.1 shows the distribution of mass numbers for fission fragments from the fission of ^{235}U. Fission yield refers to the percentage of a nuclide of a fission fragment produced per fission. Two or three free neutrons usually appear along with the fission fragments.



Source: https://www.quora.com/Why-is-promethium-produced-from-uranium-235

Fig. 7.1

The induced fission of ^{235}U may be represented by a nuclear equation of the form

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{a}_{b}P + ^{c}_{d}Q + (2 \text{ or } 3)\,^{1}_{0}n + \text{energy}$$

where P represents the lower mass number fission fragment, and Q represents the higher mass number fission fragment.

One day after the earthquake, elevated levels of Iodine-131, Cesium-134 and Cesium-137, which are common fission fragments of Uranium-235, were detected near the nuclear plant site. Fig. 7.2 shows some general information of the three fission fragments.

Isotope	**Symbol**	**Half-life**
Iodine-131	$^{131}_{53}I$	8.0197 days
Cesium-134	$^{134}_{55}Cs$	2.065 years
Cesium-137	$^{137}_{55}Cs$	30.17 years

Fig. 7.2

(a) Explain the difference between *induced* and *spontaneous* fission.

[1]

Unlike spontaneous fission, induced fission requires initial <u>neutron absorption</u> to occur. **B1**

Candidates need to answer this question by reading the passage in context.

(b) Suggest why the *percentage fission yield* in Fig. 7.1 is shown on a logarithmic scale.

[1]

Percentage fission yield varies over a wide range of many orders of magnitude. **B1**
OR
Logarithmic scale compresses the scale to accommodate the very wide range of percentage fission yield values.
OR
To enable plots on the graph to be well spaced while representing the very wide range of percentage fission yield values.

(c) Deduce the two most common ranges of mass numbers of the fission fragments (P and Q) of ^{235}U.

most common range of mass numbers of P: [1]

most common range of mass numbers of Q: [1]

From Fig. 7.1,
Most common range of mass numbers of P: **<u>90 to 100 (or 88 to 102)</u>** **A1**
Most common range of mass numbers of Q: **<u>135 to 145 (or 130 to 150)</u>** **A1**

Note that Question has already defined P as the fragment with *lower* mass. So answers for P and Q cannot swop.
Marking points:
- Correct reading of the ranges of values.
- Answers for P and Q not swopped.

(d) **(i)** Uranium-235 undergoes nuclear fission to produce Rubidium-97 and Cesium-137. Write down the nuclear equation to represent this reaction. Use the chemical symbol Rb to represent Rubidium.

[2]

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{97}_{37}Rb + ^{137}_{55}Cs + 2\,^{1}_{0}n + energy$$

- Correct value of proton number of Rb (= 37). **A1**
- Deduced that 2 neutrons are emitted. **A1**
- Deduct 1 mark for other mistakes e.g. used "=" instead of "→"; did not include $^{1}_{0}n$ on the LHS of the equation; wrong notations e.g. n^{1}_{0} or $^{2}_{0}n$ instead of $2\,^{1}_{0}n$

Note: General equation is given in the question, hence no mark purely for writing the equation.

(ii) With reference to Fig. 7.1, state the percentage fission yield of

1. Cesium-137;

2. Rubidium-97.

Indicate clearly on Fig. 7.1, the points that you read off.

percentage yield of Cesium-137= .... %

percentage yield of Rubidium-97 = ... % [1]

6% for both. (Accept also 6.5%.) **A1**

Mark for correct method (clear indication of points read on Fig. 7.1).
Accept also if answers given a 6.5% for one, and 6% for the other.

(iii) Comment on the similarity of your answers in **(d)(ii)**.

....

....

....

....

.... [1]

Percentage yields of both should be similar since **they are produced from the same (unique) fission process**. So the probabilities of both of them forming are equal/same. **B1**

(iv) State the mass number of the other fission fragment produced if one of the fragments has a mass number of 81 in the nuclear fission of Uranium-235.

mass number = ... [1]

153 or 152 **A1**

Method 1:

Using $^{235}_{92}U + ^{1}_{0}n \rightarrow ^{81}_{b}P + ^{c}_{d}Q + (2or3)\,^{1}_{0}n + energy$

By conservation of mass number,

235 + 1 = 81 + c + (2 or 3)

c = 153 or 152

Method 2:

Using Fig. 7.1,

1) Equal percentage fission yield for both. So draw a horizontal line across the “M” shaped curve in Fig. 7.1.
2) By conservation of mass number, and accounting for the neutrons present before and after the fission, it is necessary that either the two inner parts of the “M” shaped curve in Fig. 7.1 is read off, or the two outer parts.

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{81}_{b}P + ^{c}_{d}Q + (2 or 3)\,^{1}_{0}n + energy$$

236 – 81 = 155
155 – 2 = 153 (if 2 neutrons)
155 – 3 = 152 (if 3 neutrons)

(e) Deduce which isotope in Fig. 7.2 will be about 47% as radioactive after 33 years.

isotope is .... [2]

L2 **Method 1:**
47% ≈ 50%, hence **M1**
33 years is close to the half-life of the isotope.
From Fig. 7.2, the isotope with a half-life close to 33 years is **Cesium-137**. **A1**

Method 2:
$A = A_o e^{-\lambda t}$ where $\lambda = \ln 2 / t_{1/2}$

“47% as radioactive” → $A / A_o = 0.47$

$0.47 = e^{-\lambda(33 years)}$
$\lambda = 0.022879$ years^{-1}
$\ln 2 / t_{1/2} = 0.022879$ years^{-1}
$t_{1/2}$ = ~30 years
Answer: Cesium-137

(f) The release of Iodine-131, Cesium-134 and Cesium-137 into the air, ground and seawater were monitored closely by the Japanese authority because they readily decay by emitting beta-particles and gamma-rays. For example, Cesium-137 decays into Barium-137 (Ba) through beta-emission.

95% of Cesium-137 decays to a metastable excited nuclear energy state (Barium -137m) and is responsible for all emission of gamma-rays, while the other 5% directly decays to the ground state (Barium-137) of the nucleus. Fig. 7.4 illustrates the decay scheme of Cesium-137.

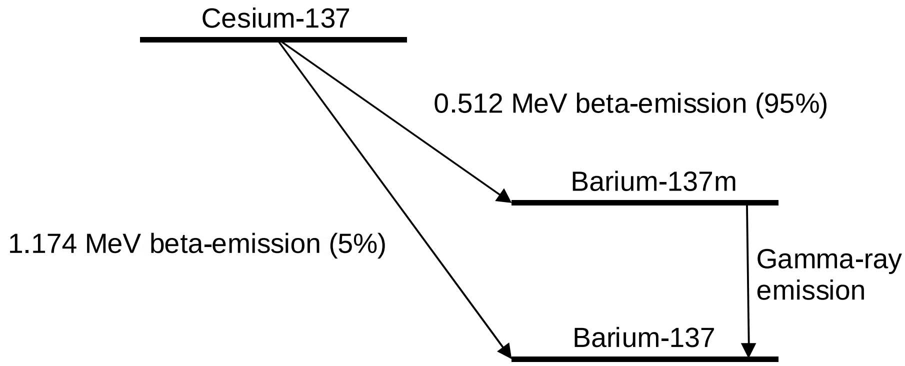


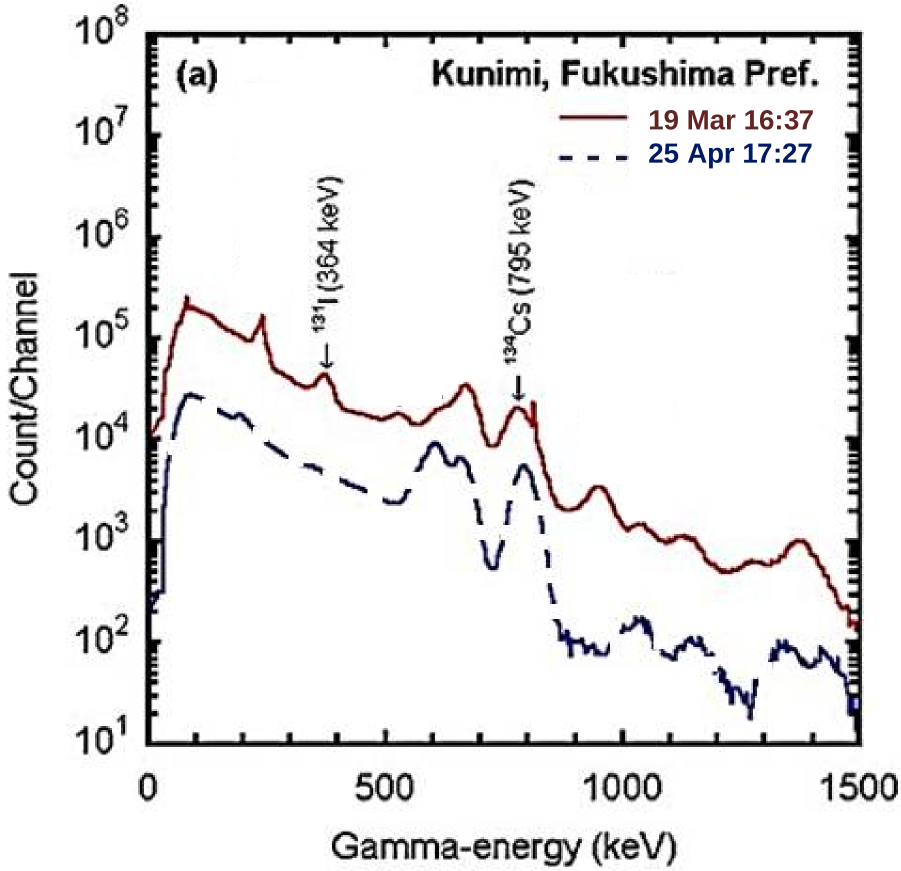
Fig. 7.4

Determine the photon energy of the gamma-rays emitted during Cesium-137 decay.

Energy of gamma-ray photon = 1.174 – 0.512 = **0.662 MeV** **A1**

energy = ... MeV [1]

(g) Gamma spectroscopy is a technique used by scientists to identify specific radioactive products from the many possible products that can be produced from the nuclear fission of Uranium-235. Fig. 7.5 shows the gamma spectra measured at Kunimi, Fukushima Prefecture on both 19 March 2011 and 25 April 2011. Photon peaks generated by Tellurium-132, Iodine-131, -132, and Cesium-134, -136 and -137 were observed in the spectra given in Fig. 7.5. The photon peaks corresponding to Iodine-131 and Cesium-134 are indicated in Fig. 7.5.



Source: https://www.nature.com/articles/srep00087 (modified)

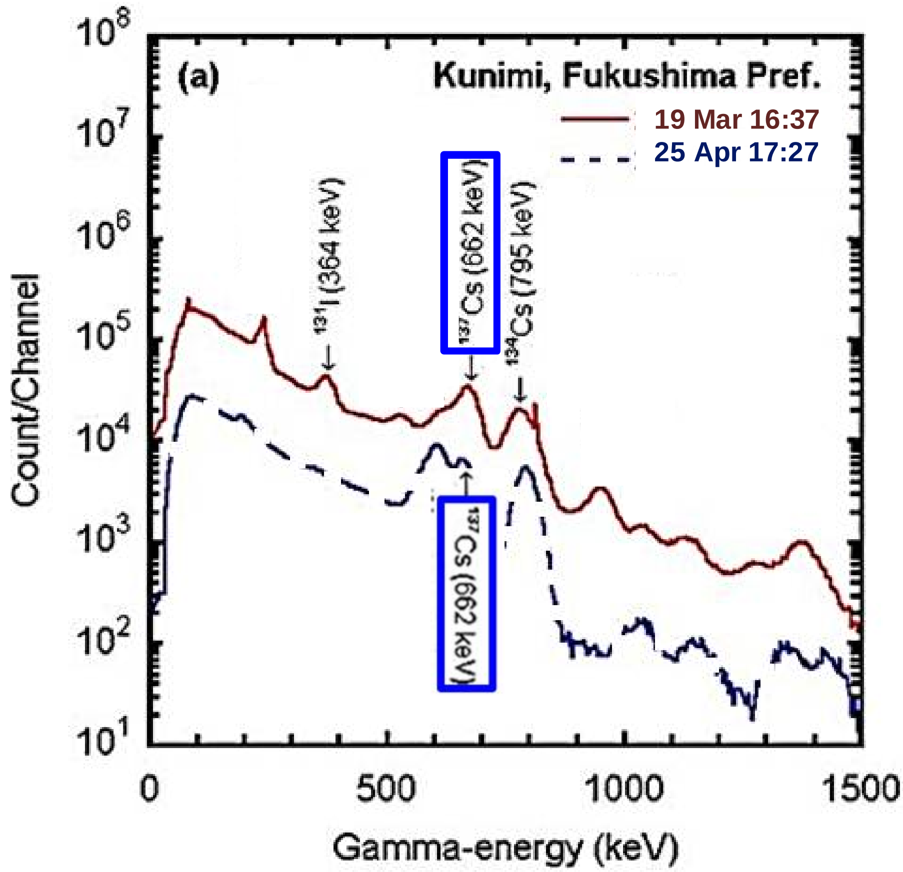
Fig. 7.5

(i) Using your answer from **(f)**, indicate on Fig. 7.5 the position of Cesium-137 in the gamma spectra. [1]

Solution: **A1**

0.662 MeV = 662 keV

Accept indication on either one of the solid or dotted line graphs.



(ii) Suggest why no photon peak of Iodine-131 was observed on 25 April 2011.

--- [1]

As seen from Fig. 7.2 , Iodine-131 has a short half-life of about 8 days, hence **a month after 19 March, on 25 April, <u>about four half-lives had passed, hence most of the iodine would have already decayed</u>**. **B1**

OR

Half life of Iodine-131 is only about 8 days, so by 25 April, yield of Iodine-131 is only 4.42% of that on 19 March which is significantly low.

[About 36 days passed → $\frac{t}{t_{\frac{1}{2}}} = \frac{36}{8} = 4.5\ half\ lives\ passed$ →

$\frac{A}{A_o} = \left(\frac{1}{2}\right)^{4.5} = 0.0442$ → A = 4.42%]

(h) The sievert (Sv) is the SI unit of effective radiation dose. It is a measure of the biological effects of radiation. One sievert carries with it a 4% chance of developing a fatal cancer in an adult, and a 0.8% chance of hereditary defects in his or her future offspring. Doses greater than one sievert received over a short time period are likely to cause radiation poisoning, possibly leading to death within weeks.

Fig. 7.6 shows the dose rate of gamma-rays recorded by a monitoring car at Fukushima Daiichi Nuclear Power Station during the incident.

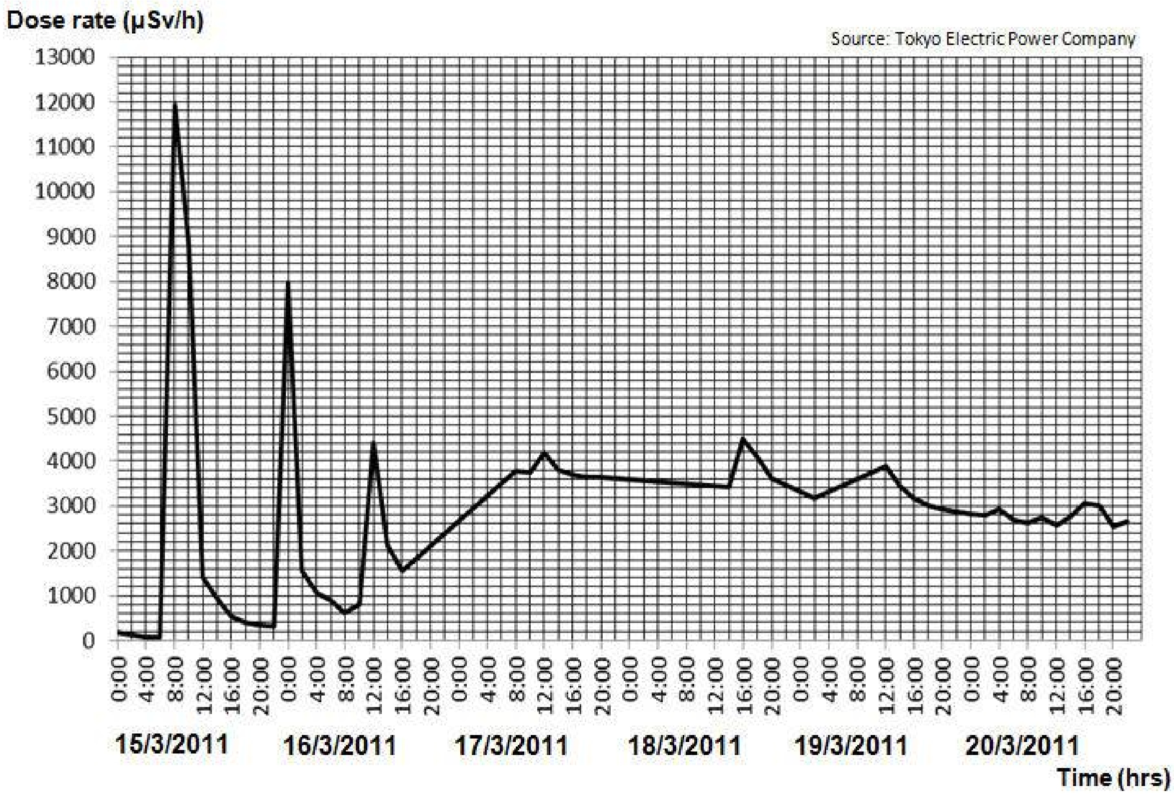


Fig. 7.6

On 15 March 2011, the Japanese Health and Labour Ministry increased the maximum permissible dose for Japanese nuclear workers from 100 mSv per year to 250 mSv per year for emergency situations.

If a worker was at the nuclear site from 16:00 hrs on 16/3/2011, estimate the time that he must be evacuated from the site before his life is in danger. Assume that the worker does not leave the site.

*Show evidence that **area under graph equals to the total dose**. **Find area equal to 250 mSv.*** **M1**

*Time should correspond to **19/3/2011 16:00 hrs**.* **A1**

Allow a range from 19/3/2011 14:00 hrs to 19/3/2011 20:00 hrs.

Accept also if students gave the total duration elapsed.
e.g. 3 days = 72 hrs

time : ------------------------------ hrs

date: ------------------------------ [2]

(i) Compare the three types of radioactive decay: alpha-decay, beta-decay and gamma-decay. Explain which of the three poses the greatest health risk if the source is:

1. outside the body;

....

....

....

....

.... [2]

Gamma-decay. Because gamma-ray is the <u>most</u> penetrating **B1**
hence <u>penetrate most deeply</u> into the body and reach living tissues. **B1**
Alpha-particle cannot penetrate the skin. Beta-particle can penetrate the skin but cannot penetrate as deeply.

2. ingested and enters the body.

....

....

....

....

.... [2]

Alpha-decay. Because alpha-particle is the <u>most</u> ionizing **B1**
hence produces the <u>most</u> number of free radicals per unit distance traveled inside the body / causes more extensive mutation of cells or DNA / highest cancer risk. **B1**

-- END OF PAPER 2 --